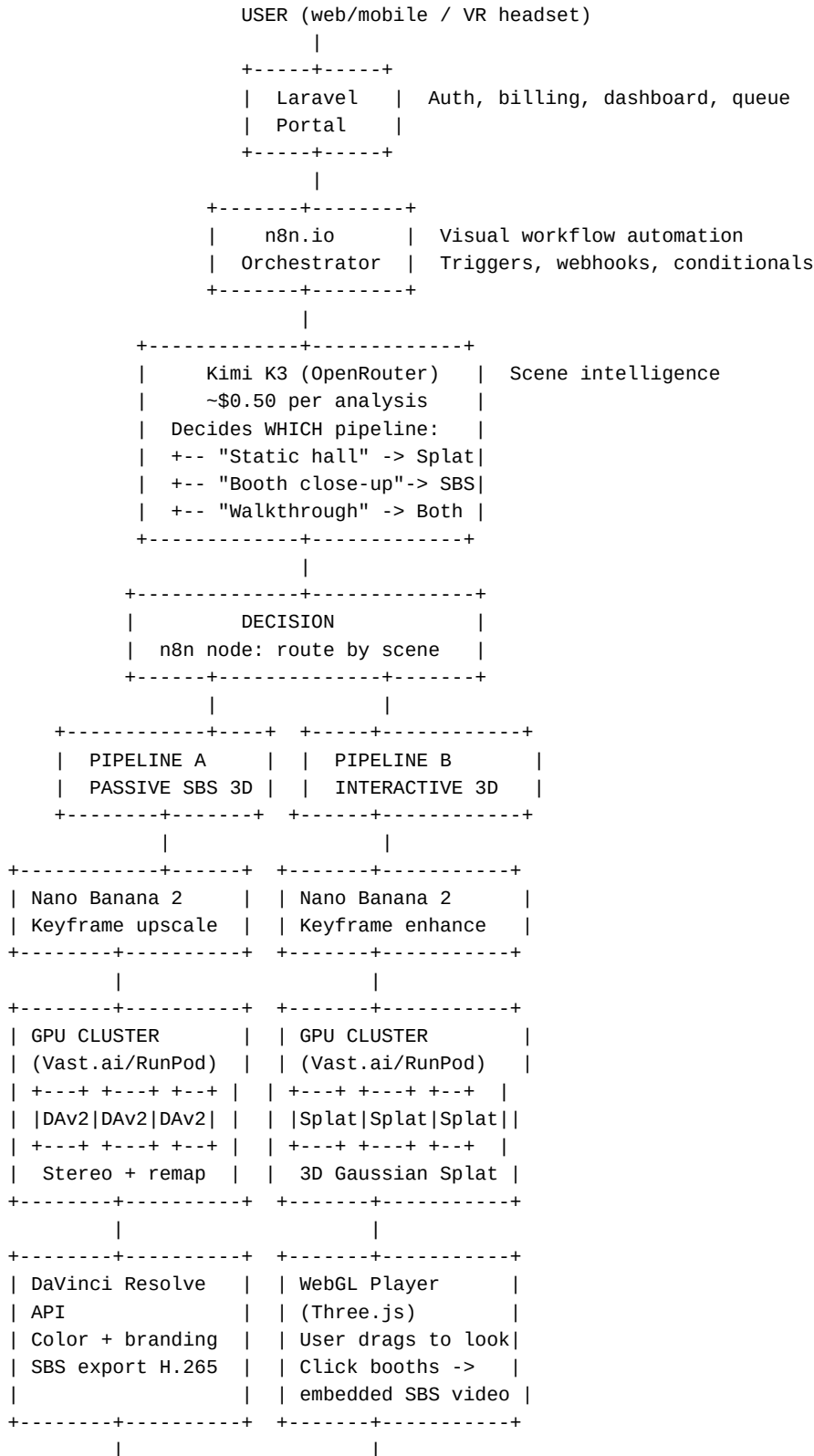
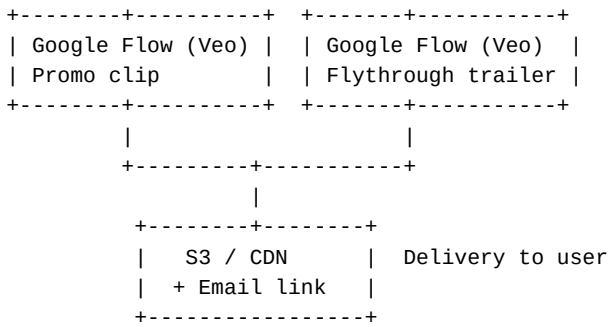


KENYA 3D PLATFORM

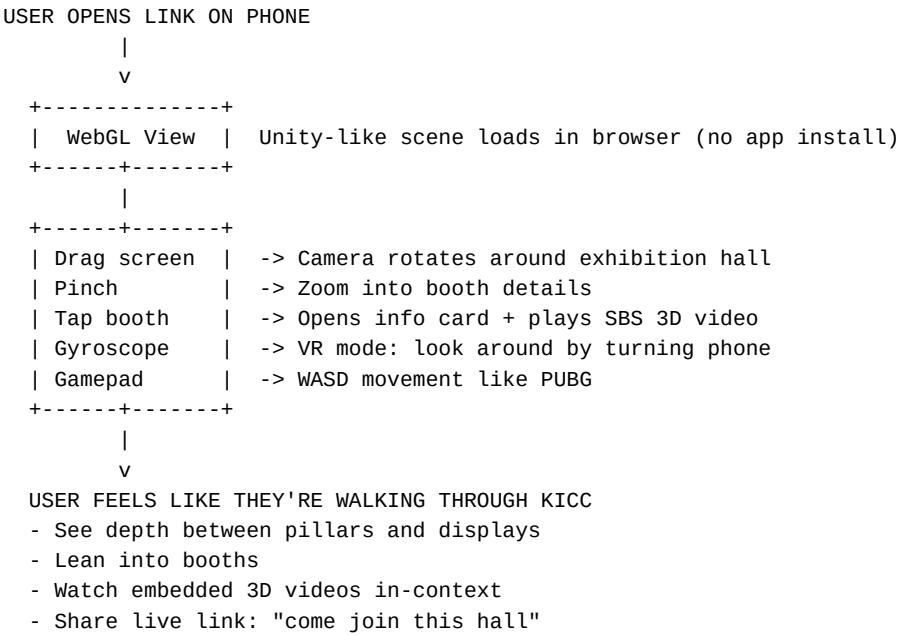
SBS 3D + Interactive Splatting + PUBG-like Control

1. COMPLETE SYSTEM ARCHITECTURE





2. THE PUBG/CALL OF DUTY INTERACTION MODEL

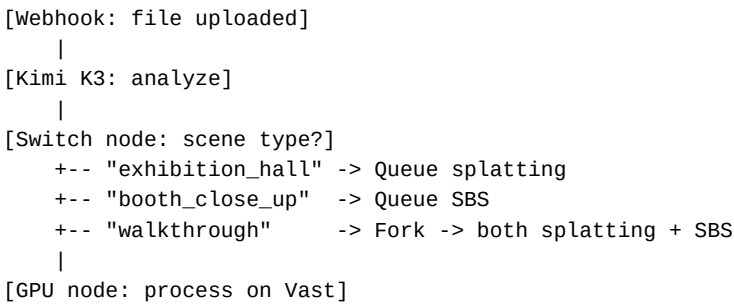


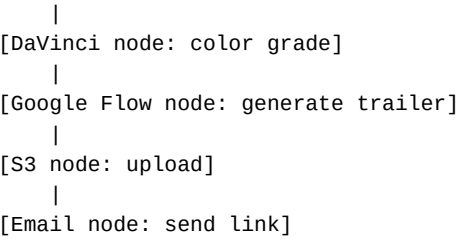
3. THE 3-TIER PRODUCT

Pricing & Delivery Tiers

Tier	Output	What User Gets	Tech	Price/min
Bronze	SBS 3D video	Watch on VR headset	DAv2 + Stereo + DaVinci	\$5
Silver	Interactive 3D link	Walk through hall on phone	Splatting + WebGL	\$15
Gold	Interactive + Embedded SBS	PUBG-like with booth videos	Splatting + SBS + Flow trailer	\$30

4. THE n8n WORKFLOW





5. PROJECT FILE TREE

```
kenya-3d-platform/
+-- laravel-backend/           User portal + API + billing
+-- n8n-workflows/             n8n JSON exports for every pipeline
|   +-- sbs-pipeline.json
|   +-- splatting-pipeline.json
|   +-- hybrid-pipeline.json
+-- pipeline/
|   +-- analyzer.py           Kimi K3 scene analysis
|   +-- depth_worker.py       DAV2 depth on GPU
|   +-- stereo_worker.py      OpenCV SBS generation
|   +-- splat_worker.py       3D Gaussian Splatting
|   +-- webgl_viewer/         Three.js interactive player
|   |   +-- index.html
|   |   +-- player.js
|   |   +-- booth-overlay.js
|   +-- refiner.py            Nano Banana 2 keyframe enhance
|   +-- resolver.py           DaVinci Resolve API
|   +-- flow_adapter.py       Google Flow API
|   +-- orchestrator.py       Master controller
+-- deploy/                   Vast.ai auto-deploy scripts
+-- opencode/                 AGENTS.md + project context
```

6. THE SOFTWARE STACK

Layer	Technology	Role
Frontend	Laravel + Vue	Upload, pay, download
AI Analysis	Kimi K3 (OpenRouter)	Scene understanding
Image Refine	Nano Banana 2 (Google)	Keyframe upscale
Depth	Depth Anything V2 (GPU)	Depth maps
Stereo	Python/OpenCV (GPU)	SBS generation
3D Reconstruction	Gaussian Splatting	Interactive 3D scenes
WebGL Viewer	Three.js / PlayCanvas	Browser 3D interaction
Post-pro	DaVinci Resolve API	Grading + branding
Promo	Google Flow (Veo API)	Marketing clips
Automation	n8n (self-hosted)	Visual workflow orchestration
Orchestration	Laravel Queues + Python	Pipeline glue
Compute	Vast.ai / RunPod	GPU rental
Storage	S3 / DigitalOcean	File delivery
Monitoring	Prometheus + Grafana	GPU, queue depth

7. SUBSCRIPTION COST SUMMARY

Service	Purpose	Monthly
OpenCode Go	AI assistant codes the system	\$10
OpenRouter (Kimi K3)	Scene intelligence per video	~\$50/100 videos
Google AI Pro	Nano Banana 2 + Google Flow	\$20
Vast.ai GPUs	Splatting + depth (as needed)	~\$500-2000
n8n (self-hosted)	Visual workflow automation	Free
Laravel Forge	Server management	\$15
DigitalOcean/S3	Storage + CDN	~\$50
DaVinci Resolve	Post-pro (one-time)	\$295 (once)

Total recurring: ~\$95/mo + GPU compute (\$500-2000)

Generated for Kenya National Exhibition Platform - Immersive 3D Content Pipeline